

Deep Learning-Based Mango Leaf Disease Detection Using Transfer Learning Models

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Abstract

Plant diseases are a major challenge in agriculture because they reduce crop yield, affect fruit quality, and cause significant economic losses to farmers. Mango is one of the most commercially important fruit crops, but it is susceptible to several fungal, bacterial, and pest-related diseases. Early and accurate disease detection can help farmers take timely preventive measures and improve agricultural productivity.

This research presents a deep learning-based approach for automatic mango leaf disease classification using transfer learning techniques. The publicly available MangoLeafBD dataset, consisting of 4,000 images divided into eight classes, was used for training and evaluation. Four pretrained convolutional neural network architectures—MobileNetV2, EfficientNetB0, ResNet50, and DenseNet121—were implemented using TensorFlow and Keras. Images were resized, normalized, and augmented before model training to improve generalization.

The performance of each model was evaluated using validation accuracy, validation loss, precision, recall, F1-score, confusion matrix, and classification report. Among the evaluated models, ResNet50 achieved the highest validation accuracy of **99.62%**, followed by EfficientNetB0 and DenseNet121 with **98.75%**, while MobileNetV2 achieved **96.25%** validation accuracy. The results demonstrate that transfer learning provides a highly efficient and accurate solution for automated mango leaf disease detection and can contribute to smart agriculture applications.

Keywords— Deep Learning, Transfer Learning, Mango Leaf Disease Detection, Image Classification, MobileNetV2, EfficientNetB0, ResNet50, DenseNet121, Precision Agriculture

1. Introduction

Agriculture is one of the most important sectors supporting food security and economic development across the world. Plant diseases significantly reduce agricultural productivity by affecting crop growth, fruit quality, and overall yield. Mango (*Mangifera indica*) is among the most widely cultivated tropical fruits and contributes substantially to the agricultural economy in many countries. However, mango cultivation is frequently threatened by diseases such as Anthracnose, Powdery Mildew, Bacterial Canker, Die Back, Gall Midge, Cutting Weevil, and Sooty Mould.

Conventional disease identification is generally performed through manual visual inspection by agricultural experts. This approach is often time-consuming, expensive, and prone to human error. In many rural regions, access to plant pathology experts is limited, making timely disease diagnosis difficult. Therefore, automated disease detection systems have become an important area of research.

Recent advances in Deep Learning, particularly Convolutional Neural Networks (CNNs), have significantly improved image classification tasks. Transfer learning enables pretrained models to adapt to new datasets with fewer training samples and reduced computational cost. In this work, four transfer learning models—MobileNetV2, EfficientNetB0, ResNet50, and DenseNet121—were implemented and compared for mango leaf disease classification using the MangoLeafBD dataset.

The objectives of this research are:

- To develop an automated mango leaf disease detection system using deep learning.
- To compare the performance of multiple transfer learning models.
- To evaluate the models using standard performance metrics.
- To identify the most suitable model for real-world agricultural applications.



Figure 1. Sample images from the MangoLeafBD dataset showing healthy and diseased mango leaves

2. Literature Review

Several studies have demonstrated the effectiveness of deep learning for plant disease detection. Ahmed et al. introduced the MangoLeafBD dataset, which contains images of healthy and diseased mango leaves collected under controlled conditions. The dataset has become a benchmark for evaluating deep learning algorithms for mango disease classification.

Transfer learning has emerged as one of the most successful approaches for agricultural image classification because pretrained models can learn meaningful image features without requiring extremely large training datasets. Models such as MobileNetV2, EfficientNetB0, ResNet50, and DenseNet121 have consistently achieved high accuracy in plant disease recognition tasks due to their robust feature extraction capabilities.

Recent research has also explored Vision Transformers (ViTs) and hybrid CNN-transformer architectures for plant disease detection. While these methods have shown promising results, CNN-based transfer learning models remain computationally efficient and highly accurate, making them suitable for practical agricultural applications.

Based on the findings of previous studies, this work performs a comparative evaluation of four popular transfer learning architectures on the MangoLeafBD dataset to determine the most effective model for mango leaf disease classification

3. Dataset Description

1. The MangoLeafBD dataset was used in this study to develop and evaluate the proposed deep learning models. It is a publicly available benchmark dataset specifically designed for mango leaf disease classification. The dataset contains a total of **4,000 RGB images** distributed equally among **eight classes**, including one healthy class and seven diseased classes.
2. Each image was captured under controlled lighting conditions with high image quality to ensure clear visualization of disease symptoms. The dataset was divided into training and validation subsets using an **80:20**

split, where 80% of the images were used for model training and 20% for validation.

- Before training, all images were resized to **224 × 224 pixels**, which is the standard input size for the pretrained convolutional neural network architectures used in this research. Data augmentation techniques such as horizontal flipping, random rotation, zooming, and brightness adjustment were applied to improve the robustness and generalization capability of the models.
- The eight disease categories available in the dataset are listed in Table I.

Table I. Mango Leaf BD Dataset Description

Class	Description
Anthraxnose	Fungal disease causing dark lesions on leaves
Bacterial Canker	Bacterial infection resulting in leaf spots and stem damage
Cutting Weevil	Insect infestation causing irregular cuts on leaves
Die Back	Drying of branches and leaves due to fungal infection
Gall Midge	Pest infestation producing abnormal swellings on leaves
Healthy	Healthy mango leaves without disease symptoms
Powdery Mildew	White powder-like fungal growth on leaf surface
Sooty Mould	Black fungal coating affecting photosynthesis

Anthraxnose



Bacterial Canker



Cutting Weevil



Die Back



Gall Midge



Healthy



Powdery Mildew



Sooty Mould



Figure 2. Representative images from the MangoLeafBD dataset showing healthy and diseased mango leaves belonging to eight different classes.

4. Proposed Methodology

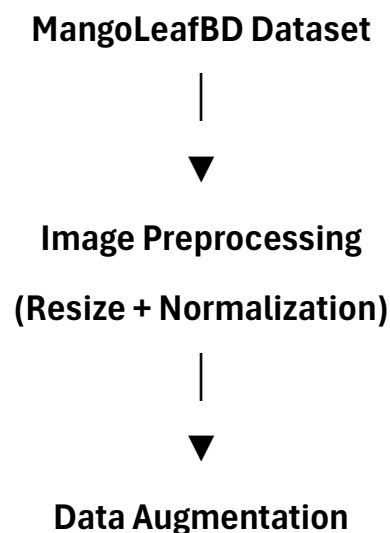
The proposed methodology consists of a sequence of image preprocessing, transfer learning, model training, and performance evaluation stages. The complete workflow adopted in this study is illustrated in Fig. 3.

Initially, the MangoLeafBD dataset was imported into the Google Colab environment. The images were resized to **224 × 224 pixels** and normalized before training. Data augmentation techniques were applied to increase dataset diversity and minimize overfitting.

Four pretrained transfer learning models—**MobileNetV2, EfficientNetB0, ResNet50, and DenseNet121**—were selected because of their proven performance in image classification tasks. The convolutional base of each model was initialized with ImageNet weights, while the final classification layer was modified to classify the eight mango leaf categories.

The models were trained using the Adam optimizer and Sparse Categorical Crossentropy loss function. Early stopping and learning rate reduction callbacks were employed to prevent overfitting and improve convergence. After training, each model was evaluated using validation accuracy, validation loss, precision, recall, F1-score, confusion matrix, and classification report.

The performance of all four models was compared to determine the most effective architecture for mango leaf disease classification.



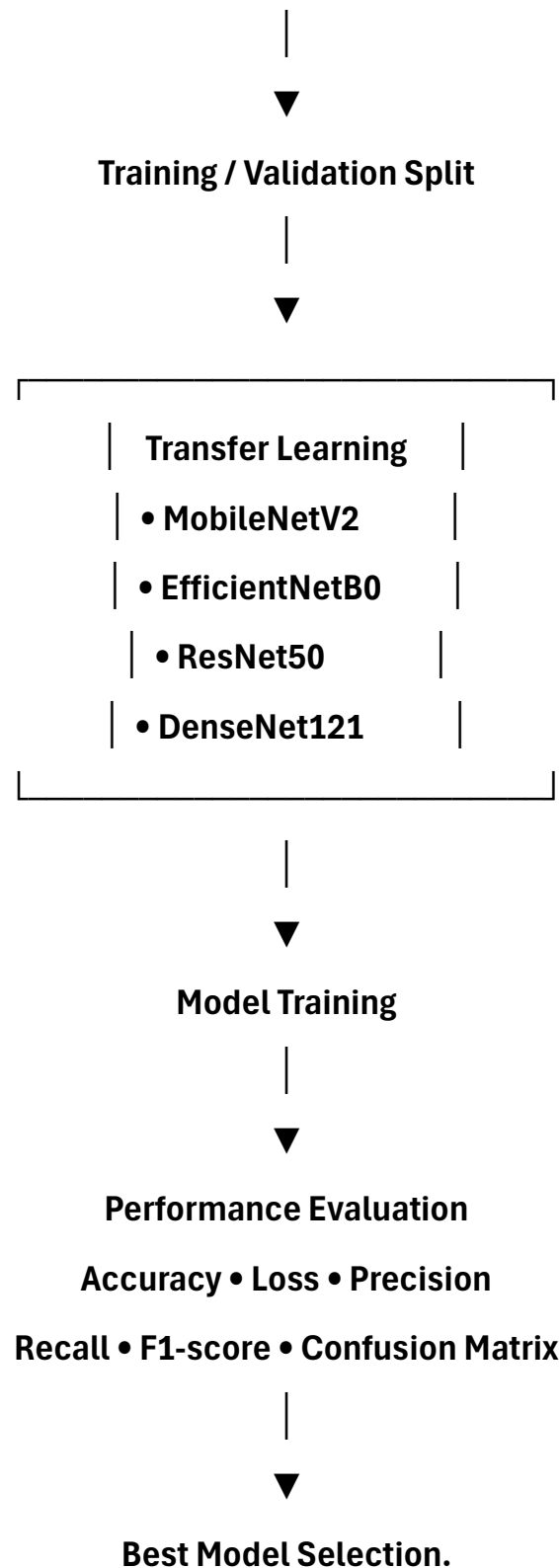


Figure 3. Proposed workflow of the deep learning-based mango leaf disease detection system

5. Experimental Setup

The experiments were carried out using **Google Colab** with NVIDIA T4 GPU acceleration. The deep learning models were implemented using **Python**, **TensorFlow**, and **Keras**. Additional libraries including NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn were used for data processing, visualization, and evaluation.

All images were resized to **224 × 224 pixels** before being fed into the pretrained networks. An 80:20 train-validation split was adopted for model training and evaluation. The Adam optimizer with a learning rate of **0.001** was used for optimization, while Sparse Categorical Crossentropy served as the loss function. Early stopping and learning rate reduction callbacks were incorporated to improve convergence and reduce overfitting.

The evaluation metrics used in this work include:

- Validation Accuracy
- Validation Loss
- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report

6. Results and Discussion

The proposed deep learning framework was evaluated using four pretrained transfer learning models: **MobileNetV2**, **EfficientNetB0**, **ResNet50**, and **DenseNet121**. The models were trained and validated using the MangoLeafBD dataset. Their performance was assessed using validation accuracy, validation loss, precision, recall, F1-score, confusion matrix, and classification report.

The experimental results indicate that all four models achieved high classification performance, demonstrating the effectiveness of transfer learning for mango leaf disease detection. Among the evaluated architectures, **ResNet50** achieved the highest validation accuracy of **99.62%** with the lowest validation loss of **0.0482**, making it the best-performing model in this study.

EfficientNetB0 and DenseNet121 also achieved excellent validation accuracy of **98.75%**, while MobileNetV2 achieved **96.25%** accuracy with comparatively higher validation loss.

The confusion matrix and classification reports show that the models correctly classified most of the mango leaf disease categories with very few misclassifications. This demonstrates that the pretrained convolutional neural network models were able to learn discriminative image features effectively despite the relatively small dataset.

Overall, the experimental results confirm that transfer learning significantly improves classification performance while reducing computational cost and training time. Therefore, these models can be effectively utilized for real-world agricultural disease diagnosis and smart farming applications.

Table II. Performance Comparison of Transfer Learning Models

Model	Validation Accuracy (%)	Validation Loss
MobileNetV2	96.25	0.1847
EfficientNetB0	98.75	0.0788
ResNet50	99.62	0.0482
DenseNet121	98.75	0.0938

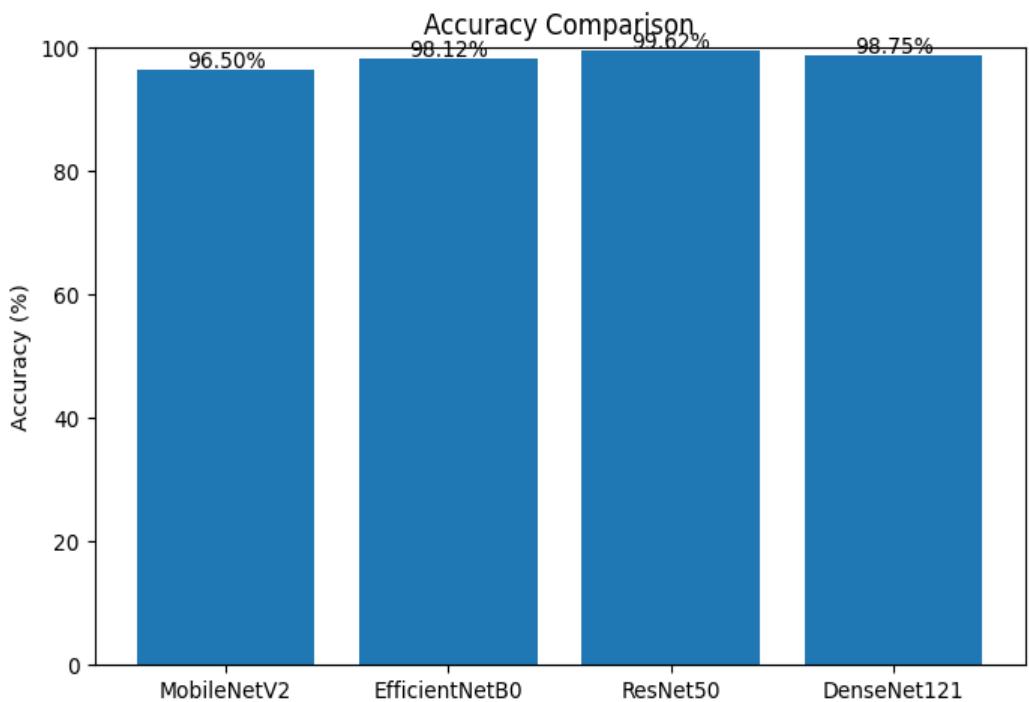


Figure 4. Validation accuracy comparison of the four transfer learning models.

Confusion Matrix of ResNet50

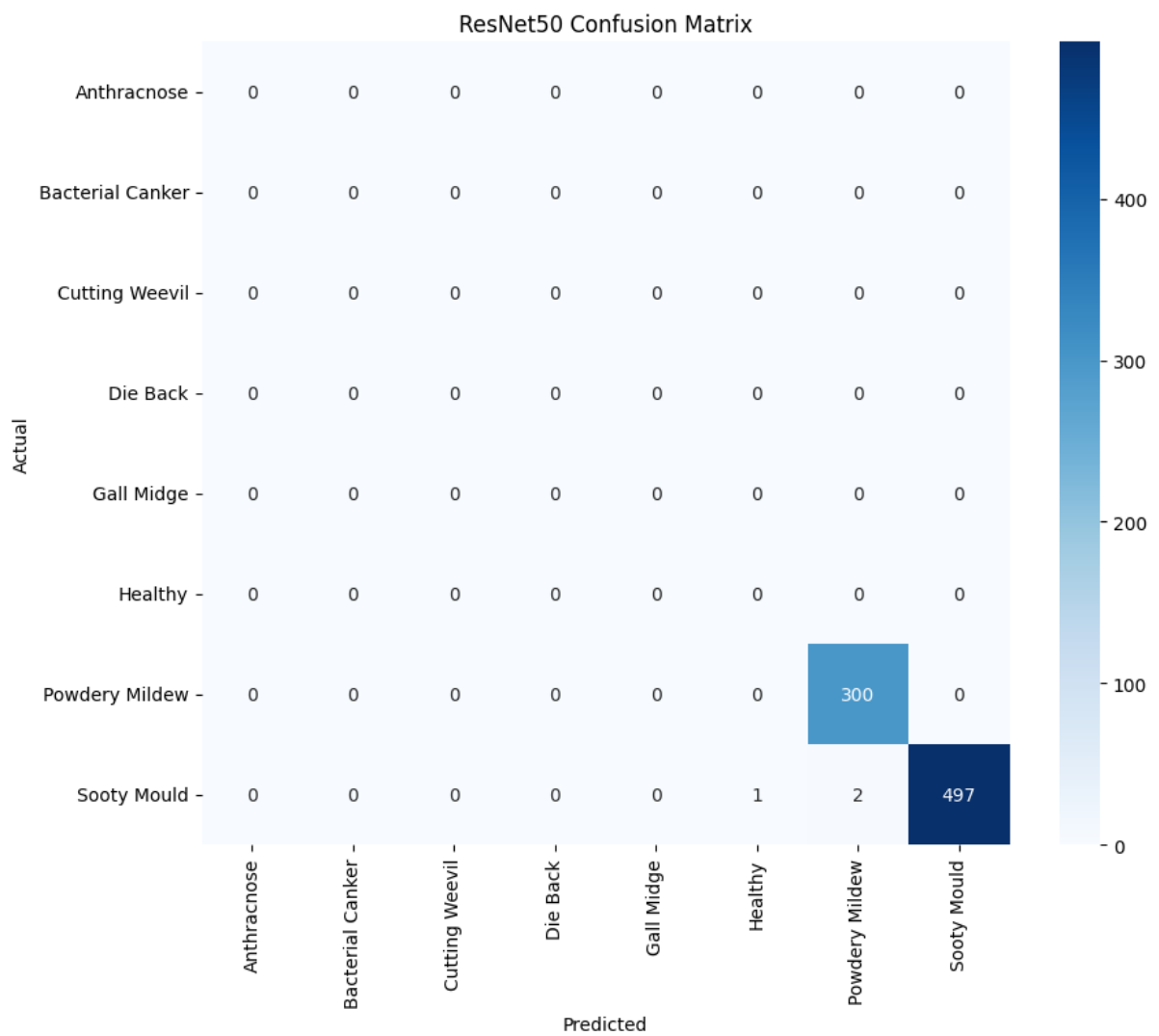


Figure 5. Confusion matrix of the ResNet50 model for mango leaf disease classification.

7.Conclusion

This research presented a deep learning-based approach for automatic mango leaf disease detection using transfer learning. Four pretrained convolutional neural network models—MobileNetV2, EfficientNetB0, ResNet50, and DenseNet121—were implemented and evaluated using the MangoLeafBD dataset. Image preprocessing, data augmentation, and

transfer learning techniques were employed to improve model performance and reduce training time. Experimental evaluation demonstrated that all four models achieved high classification accuracy. Among them, **ResNet50** achieved the best overall performance with a validation accuracy of **99.62%** and the lowest validation loss of **0.0482**. EfficientNetB0 and DenseNet121 also produced excellent results, while MobileNetV2 provided competitive performance with lower computational complexity. The results confirm that transfer learning is an effective approach for automated plant disease detection and has significant potential for deployment in smart agriculture. The proposed system can assist farmers and agricultural experts in early disease identification, enabling timely treatment and reducing crop losses.

8. Future Scope

The proposed work can be extended in several directions to further improve disease detection accuracy and practical usability. Future research may include collecting a larger and more diverse dataset captured under real field conditions, integrating Vision Transformer (ViT) architectures and attention mechanisms, and developing lightweight models suitable for deployment on mobile devices. Additionally, the proposed system can be integrated with IoT-based smart farming solutions to provide real-time disease monitoring and automated recommendations for disease management.

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